

Long-lived memory in sliding spin chains

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In this work, we do cool shit.

INTRODUCTION

Memory in physical systems is interesting.

Memory in 1d is especially interesting, and poorly understood.

In this paper, we consider spin systems that retain a single bit of memory in the sign of their magnetization. We consequently define the memory time t_{mem} as

$$t_{\text{mem}} = \mathbb{E} \min_{s=\pm 1} \{t : |m(t)| < 1/2, m(0) = s\} \quad (1)$$

where the constant $1/2$ is chosen rather arbitrarily. t_{mem} sets a lower bound on the mixing time.

Consider the case of Glauber dynamics run at temperature T for the 1d Ising chain with coupling J . Once a minority domain (of any size) is formed, its endpoints will execute unbiased random walks. Since the energy cost for a minority domain is $\Delta E = 4J$, it is thus easy to show that $t_{\text{mem}} < be^{4\beta J}$, where b is a constant. More generally, for *any* classical system undergoing dynamics obeying detailed balance with respect to a local Hamiltonian H , general physical arguments show $t_{\text{mem}} < be^{c\beta J}$, where J is the largest local energy scale in H , and c is another constant.

In this paper, we show that subjecting a ferromagnetic 1d Ising model on a two-leg ladder to a very simple type of nonequilibrium drive can significantly increase t_{mem} . We drive the system out of equilibrium by sliding the two legs of the ladder past one another at a relative velocity v , as illustrated in Fig. 1 a. More precisely, letting $s_i^{(t/b)} \in \{\pm 1\}$ be the i th spin on the top / bottom leg of the ladder, we consider Glauber dynamics at temperature T under the time-dependent Hamiltonian [?]]

$$H(t) = -J \sum_{i;a \in \{t,b\}} s_i^{(a)} s_{i+1}^{(a)} - J \sum_i s_{[i-vt/2]}^{(t)} s_{[i+vt/2]}^{(b)}. \quad (2)$$

Let us work in units of time where each spin experiences on average one Glauber update per unit time. When $v = 0$, the dynamics is simply that of a static two-leg Ising chain, and $t_{\text{mem}} \sim e^{c\beta J}$ for constant c . The main result of this work is to show that t_{mem} is significantly increased when v is greater than a constant threshold. In particular, we give analytic arguments—and verify in numerics—that as long as $v > 2$,

$$t_{\text{mem}} \gtrsim be^{4(\beta J)^2 v (\ln v + c)} \quad (3)$$

where b, c are positive constants that are independent of $v, \beta J$ in the limits $v, \beta J \gg 1$. While t_{mem} is finite in the

thermodynamic limit for all T , it rapidly gets astronomically large as $T \rightarrow 0$. For example, when $T = J/2$ and $v = 3$, we find numerically that $t_{\text{mem}} \sim 10^{50}$ (see below for more details). The remainder of this paper is dedicated to understanding and expanding upon this result.

UNDERSTANDING THE MECHANISM

To understand why sliding increases t_{mem} at large enough v , consider what happens to minority domains under this dynamics. A minority domain on only a single chain is eroded ballistically from its endpoints. From the transition rates of Glauber dynamics, the endpoints of a single-chain minority domain contract at velocity

$$\delta = (1 + e^{-2\beta J})^{-1}. \quad (4)$$

This is true even when $v = 0$.

A minority domain that spans *both* chains is more problematic. When $v = 0$, its endpoints diffuse. However, when $v > 0$, the sliding dynamics attempts to shear the domain, ripping it apart into two spatially-separated single-chain minority domains. The Glauber dynamics attempts to arrest this shearing, and it is easy to check that this succeeds if $v/2 < \delta$, in which case the domains on both chains stick together. When $v/2 > \delta$ however, the domains are sheared apart faster than they try to stick together.[?]] A visualization of this process is shown in Fig. 1 b.

How effective is this shearing mechanism? First consider what happens when $v \rightarrow \infty$: here, domains are sheared apart instantly, and a spin on one chain couples to spins on the other chain in a random order. The inter-chain interactions effectively become all-to-all, and the model enters a mean field limit, where it is easy to show the existence of a phase with long range order. In this regime, a system of size L has $\lim_{L \rightarrow \infty} t_{\text{mem}} = \infty$ at low enough T , and in fact we only need let v diverge as $\Omega(\log L)$ for this to be true.

What about at finite v ?

THE CRITICAL LENGTHSCALE

Let us refer to the spacetime region where spin flips in the metastable region can prevent the $m = -2$ region of the minority domain from eroding as the “danger region” $\text{DR}(l)$ (c.f. Fig. 2). Then ξ_{er} can be estimated as the size

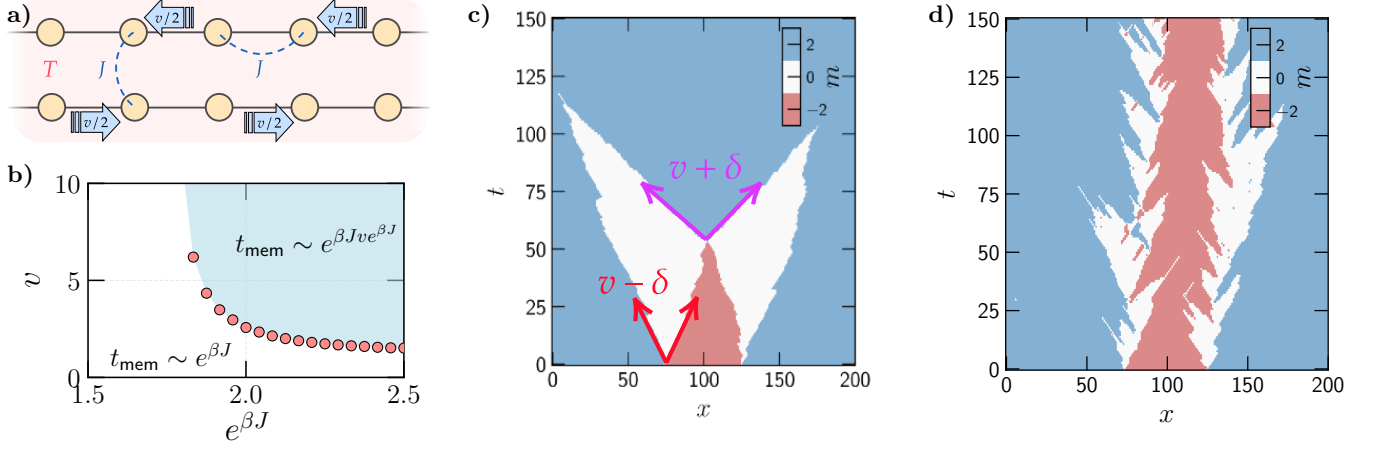


FIG. 1. **a)** Schematic of the dynamics. Two ferromagnetically coupled ferromagnetic 1d Ising chains slide past one another at relative velocity v , and both couple to a bath at temperature T . **b)** Dynamic phase diagram in the $e^{\beta J}$ - v plane. In the blue shaded region, $t_{\text{mem}} \sim e^{\beta J v e^{\beta J}}$, while in the white region the scaling crosses over to the trivial $t_{\text{mem}} \sim e^{\beta J}$ (ethan: need constants in the exponents etc.). The pink points are drawn by fixing $e^{\beta J}$, and then finding the smallest value of v such that $t_{\text{mem}}(v) > 10t_{\text{mem}}(0)$. **c)** A spacetime history of the magnetization, with $\beta = \infty, v = 2$. The indicated purple and red arrows denote interfaces that move with expected speeds $v + \delta$ and $v - \delta$, respectively. **d)** A history with $\beta = 2, v = 2$, with the minority domain in the initial state having size $l > \xi_{\text{er}}$.

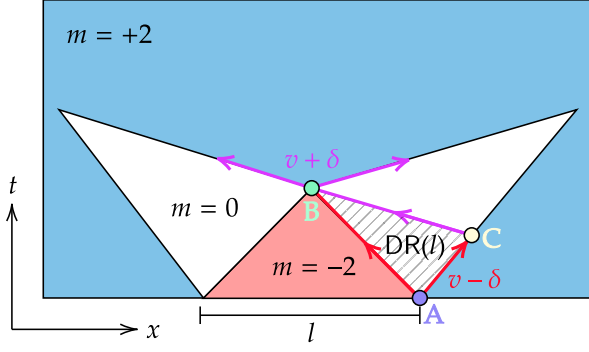


FIG. 2. How a minority domain of size l is eroded under ideal sliding dynamics at $\beta \rightarrow \infty$. The white regions have zero magnetization, and are metastable. The red arrows indicate interfaces that move with expected speed $v - \delta$, and the purple arrows those with speed $v + \delta$. The danger region $\text{DR}(l)$ indicated by the shaded triangle is the spacetime region where spin flips in the metastable zero-magnetization area can propagate to prevent the erosion of the $m = -2$ (red) part of the initial minority domain.

l where getting an error somewhere in $\text{DR}(l)$ becomes likely:

$$e^{-2\beta J} |\text{DR}(\xi_{\text{er}})| \sim 1. \quad (5)$$

$|\text{DR}(l)|$ can be determined from simple geometric considerations (c.f. fig. 1 c) as $|\text{DR}(l)| \approx l^2 \delta / (4v(v - \delta))$ [1]. From (5), we then have

$$\xi_{\text{er}} = c v e^{\beta J}, \quad (6)$$

where c is a constant independent of v to leading order in δ/v .

To determine t_{mem} , we thus need to understand the most likely process by which fluctuations produce droplets of size ξ_{er} . If such droplets had to be created “all at once” by rare thermal fluctuations, or had to be built up continuously from a single minority domain that expands from size 1 to size ξ_{er} , we would expect $t_{\text{mem}} \sim (e^{4\beta J})^{\xi_{\text{er}}}$. This estimate is too naive however, as there is a more efficient way of building up minority domains.

To understand this mechanism, consult the schematic of minority domain erosion in Fig. 2. If the correct type of spin flip happens in DR (in this case, a spin in DR flipping from $+1$ to -1), it can increase the size of the minority domain (c.f. Fig. 1 d). Letting $x_{A/C}$ be the spatial coordinates of the points **A**, **C** in Fig. 2, the largest amount by which such an event can extend the size of the minority domain is $x_C - x_A$, which one may easily show [1] equals $l\delta/2v$. A spin flip event with likelihood $\sim e^{-2\beta J}$ can therefore increase the size of the minority domain as $l \rightarrow l(1 + \delta/2v)$. Since the domain size increases multiplicatively, the number $n_{\xi_{\text{er}}}$ of spin flips needed to grow the domain to size ξ_{er} is $n_{\xi_{\text{er}}} = \ln(\xi_{\text{er}})/\ln(1 + \delta/2v)$. We therefore expect

$$t_{\text{mem}} \gtrsim b(e^{2\beta J})^{\ln(\xi_{\text{er}})/\ln(1+\delta/2v)} \sim b e^{2(\beta J)^2/\ln(1+\delta/2v)}, \quad (7)$$

for a constant $b > 0$. Expanding in $\delta/2v \ll 1$ then gives the main result (3).

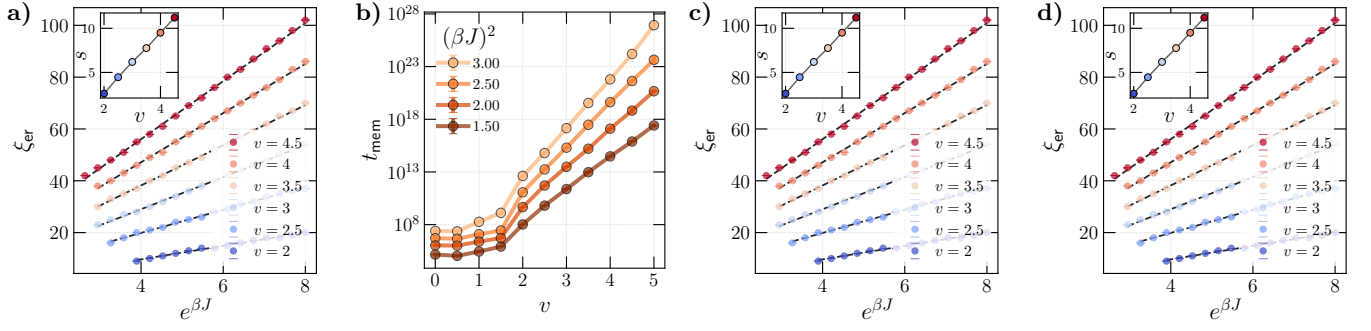


FIG. 3. **a)** ξ_{er} against $e^{\beta J}$ for different values of v . The inset shows the slopes s of linear fits of ξ_{er} to $e^{\beta J}$ (drawn as dashed black lines), confirming (6). **b)** Estimates for t_{mem} as a function of v , obtained with forward flux sampling. **c)** t_{mem} plotted against $(\beta J)^2$. **d)** $E(v)/E(0)$. At low enough temperatures, the energy decreases at large enough v .

CONFIRMING THE MEMORY TIME

We now confirm these predictions in numerics. We start with our prediction for ξ_{er} , which we estimate as follows. First, we prepare an initial state containing a minority domain of length l in a system of size $L = 5vl$. We then evolve for a time l , and determine whether or not the total number of minority spins has decreased to less than l . When then look for the smallest value of l such that the probability $P_{\text{er}}(l)$ of this happening is less than $3/4$, thereby defining ξ_{er} as

$$\xi_{\text{er}} = \min\{l : P_{\text{er}}(l) > 3/4\}. \quad (8)$$

Fig. 3 **a** shows the result of measuring (8) in numerics, $e^{\beta J}$, confirming (6).

We now confirm the scaling of the memory time in (3). This is infeasible with direct Monte Carlo updates, but fortunately forward flux sampling [6] allows us to get estimates for the memory time even when t_{mem} is extremely large (for details about the numerics, see [1]).

Fig. 3 **b** and **c** show t_{mem} plotted against v and $(\beta J)^2$, respectively.

FRICTIONAL COOLING

Let $E(v)$ denote the expected average energy at velocity v .

OUTLOOK

This work has shown that a very simple nonequilibrium drive can dramatically slow down relaxation in a locally-

interacting 1d spin chain.

Future work, as well as extending this mechanism to increase the lifetime of *quantum* information.

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- [1] See the following github repository for the supplementary material and full codebase: github.com/ethanlake/sliding-layers (2026).
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